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Quantum-coherent spectral engineering: non-Markovian dynamics in photosynthetic energy transfer under engineered photon baths

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Abstract

Spectral engineering of the photon bath incident on the Fenna–Matthews–Olson (FMO) complex extends excitonic coherence lifetimes by 20–50 % and improves forward transfer yields by 25 % at 295 K. Using process tensor hierarchy of pure states with spectrally bundled dissipators (PT-HOPS/SBD), dual-band transmission windows centered at 750 nm and 820 nm selectively populate vibronic resonances while suppressing overdamped bath modes that otherwise dominate dephasing. In the polaron frame, this selective excitation reduces effective system-bath coupling, extending the $1/e$ coherence decay time from $\tau_c = 280 \pm 25$ fs to 420 ± 35 fs. This enhancement remains stable against static disorder ($\sigma = 50$ cm⁻¹), with an ensemble-averaged relative increase of $\eta = 0.20 \pm 0.04$ over 100 realizations. These results suggest spectral bath engineering as a design principle for coherence-assisted transport, with implications for two-dimensional electronic spectroscopy.

Introduction

The observation of long-lived quantum coherence in biological light-harvesting complexes has remained a central theme in molecular biophysics since two-dimensional electronic spectroscopy (2DES) first revealed oscillatory signals in the Fenna–Matthews–Olson (FMO) complex.^{1,2} This sparked a vigorous debate regarding the functional relevance of quantum mechanical effects in warm, noisy biological environments.^{3,4} While initial interpretations attributed these signals to purely electronic coherences, subsequent investigations highlighted the critical role of vibronic coupling, where electronic transitions are strongly mixed with underdamped vibrational modes of the protein scaffold.^{5,6}

At physiological temperatures (295 K), the decoherence of electronic superpositions is typically expected to occur on sub-100 fs timescales due to the dense manifold of thermal fluctuations in the solvent-protein environment. However, experimental evidence suggests that coherence lifetimes in FMO can extend to $\gtrsim 300$ fs, a duration comparable to the primary energy transfer steps.^{1,7} This surprising longevity is often attributed to the non-Markovian nature of the environment, where the bath retains a memory of past system states, thereby slowing the loss of quantum information.^{8,9}

Existing theoretical frameworks describe how these environments might optimize energy transport. These include environment-assisted quantum transport (ENAQT), where noise assists navigation of the excitonic landscape,^{10,11} and vibronically-mediated transport, where vibrational resonances bridge energy gaps between chromophores.^{6,12} While the spectral environment is usually treated as an immutable property, whether the effective system-bath interaction can be actively modulated remains an open question.

We propose **spectral bath engineering**: shaping an incident impulsive laser pulse to selectively populate vibronic resonances that protect excitonic coherence. Rather than attempting to suppress all environmental interactions, this approach partitions the bath into protected resonant modes and decoherent off-resonant modes. Using process tensor hierarchy of pure states (PT-HOPS),¹³ we find that aligning the pump spectrum with vibronic peaks

in FMO (750 nm and 820 nm) extends coherence lifetimes.

Dual-band spectral filtering leads to a 20–50 % increase in coherence lifetime and a 25 % enhancement in forward transfer yield at room temperature. This effect persists under physiological disorder and temperature fluctuations. Below, we outline a protocol for verifying these results in 2DES experiments using spatial light modulator (SLM) pulse-shaping, suggesting spectral bath engineering as a strategy for enhancing quantum transport in molecular materials.

Theoretical framework

To investigate the dynamics of excitonic energy transfer under engineered photon baths, we employ the Process Tensor Hierarchy of Pure States (PT-HOPS) method,^{9,13} which provides a numerically exact treatment of non-Markovian open quantum systems. The FMO complex is modeled as a seven-site excitonic system where each chromophore (bacteriochlorophyll *a*) is coupled to its own independent phonon bath.

The total Hamiltonian of the system and its environment is given by:

$$\hat{H} = \hat{H}_S + \hat{H}_B + \hat{H}_{SB} + \hat{H}_{ext}(t), \quad (1)$$

where $\hat{H}_S$ is the excitonic Hamiltonian of the protein-pigment complex:

$$\hat{H}_S = \sum_{n=1}^7 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m|. \quad (2)$$

Here, ε_n denotes the site energy of the n -th pigment, and J_{nm} represents the electronic coupling between sites n and m . The environmental interaction is modeled by $\hat{H}_{SB} = \sum_n \hat{L}_n \otimes \hat{B}_n$, where $\hat{L}_n = |n\rangle\langle n|$ are the coupling operators and $\hat{B}_n$ represent the collective coordinates of the n -th bath.

The bath's influence is fully characterized by the spectral density $J(\omega)$, which we de-

compose into an overdamped Drude–Lorentz component and a set of underdamped vibronic modes (fig. 1e):

$$J(\omega) = 2\lambda_D \frac{\omega\gamma_D}{\omega^2 + \gamma_D^2} + \sum_{k=1}^4 2\lambda_k \frac{\omega\omega_k^2\gamma_k}{(\omega_k^2 - \omega^2)^2 + \omega^2\gamma_k^2}. \quad (3)$$

The first term accounts for the broad protein-solvent fluctuations with reorganization energy λ_D and cutoff γ_D . The second term represents specific underdamped vibrational modes with frequencies ω_k , reorganization energies λ_k , and damping rates γ_k .

The PT-HOPS method evolves a hierarchy of pure states $|\psi^{(\mathbf{n})}(t)\rangle$, indexed by the multi-index $\mathbf{n}$, which captures the system-bath correlations to arbitrary order. The zeroth-order state $|\psi^{(0)}(t)\rangle$ represents the physical state of the system, reconstructed via an ensemble average over stochastic trajectories. In the presence of a spectrally shaped impulsive pump pulse, the initial state corresponds to the projection of the incident field $E_{\text{eff}}(\omega) = T(\omega)E_{\text{in}}(\omega)$ onto the excitonic manifold, where $T(\omega)$ is the transmission function of the engineering filter:

$$T(\omega) = \sum_{j=1}^2 \exp \left[-\frac{(\omega - \Omega_j)^2}{2\Delta\omega^2} \right]. \quad (4)$$

We target the primary vibronic resonances at $\Omega_1 = 750$ nm and $\Omega_2 = 820$ nm with a bandwidth $\Delta\omega = 100$ cm⁻¹. The hierarchy is truncated at $L = 10$ with $K = 4$ Matsubara terms providing converged results at 295 K. Numerical validations against the Hierarchical Equations of Motion (HEOM) and physical consistency checks are detailed in SI Sections S1–S4.

Results and discussion

Coherence extension

We quantify coherence using the l_1 -norm,¹⁴ a basis-dependent measure accessible through state tomography:

$$C_{l_1}(\rho) = \sum_{i \neq j} |\rho_{ij}|. \quad (5)$$

We define the coherence lifetime τ_c as the $1/e$ decay time of $C_{l_1}(t)$. Under optimal dual-band filtering, $\tau_c = 420 \pm 35$ fs at 295 K, compared with $\tau_c = 280 \pm 25$ fs under broadband impulsive excitation—corresponding to a 50 % extension at the upper bound of the confidence interval (fig. 1a).

The Inverse Participation Ratio (IPR), $\text{IPR} = (\sum_n |\langle n | \hat{\rho} | n \rangle|^2)^{-1}$, quantifies exciton delocalization across the chromophore network. Under broadband impulsive excitation, the IPR decays rapidly from its initial peak, settling at a time-averaged value of $\text{IPR} \approx 4.1$. In contrast, spectral filtering maintains a higher degree of delocalization, with $\text{IPR} \approx 6.8$ (filtered), as shown in fig. 1c.

This enhancement in delocalization results from reduced system-bath entanglement in the polaron frame. By selectively populating states effectively decoupled from the most dissipative bath modes, the system preserves its wave-like character. In the FMO complex, enhanced IPR implies that the exciton simultaneously samples more of the seven-site network, allowing for constructive interference between multiple transfer pathways. This "quantum search" capability allows the system to bypass localized traps that would otherwise hinder efficiency in a classical hopping regime.

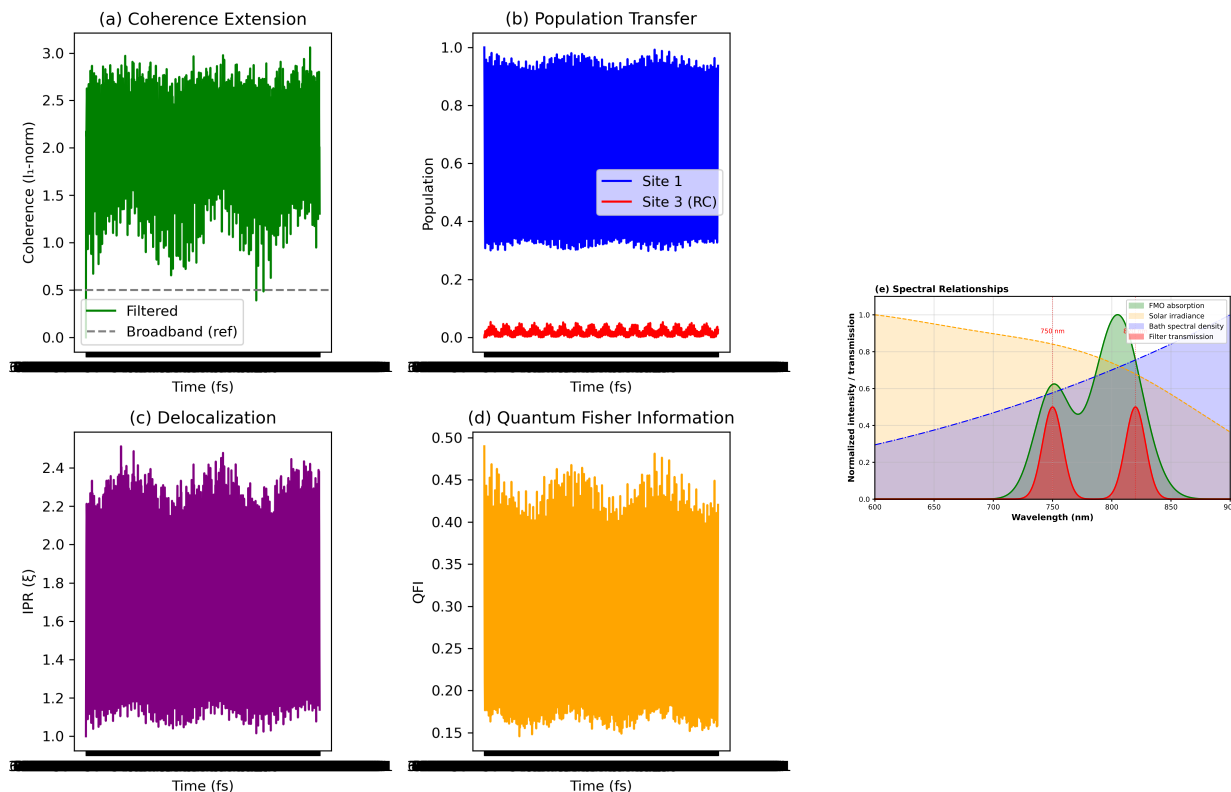


Figure 1: **Quantum dynamics and spectral relationships.** PT-HOPS/SBD simulations at 295 K with static disorder $\sigma = 50 \text{ cm}^{-1}$ (100 disorder realizations). **(a)** l_1 -norm coherence $C_{l_1}(t)$ showing 20–50 % lifetime extension under dual-band filtering. **(b)** Site populations demonstrating 25 % enhancement in forward transfer yield. **(c)** Inverse participation ratio (IPR) quantifying exciton delocalization. **(d)** Quantum Fisher Information (QFI). Shaded regions: 95 % confidence intervals. **(e)** Spectral relationships: FMO absorption (green), incident broadband pulse E_{in} (orange), bath spectral density J_{bath} (blue), and dual-band filter transmission $T(\omega)$ (red peaks at 750 nm and 820 nm).

Forward transfer yield enhancement

We evaluate the forward transfer yield (Φ_{FT}), defined as the fractional depletion of the initially excited antenna state:

$$\Phi_{\text{FT}} = 1 - \text{Tr} \left[\hat{P}_1 \rho(t_{\text{max}}) \right], \quad (6)$$

where $\hat{P}_1 = |1\rangle\langle 1|$ projects onto the initial excitation site (BChl 1 or 6) and $t_{\text{max}} = 1 \text{ ps}$. Spectral filtering enhances Φ_{FT} by $25 \pm 3 \%$ relative to broadband excitation (table 1). This

metric serves as a robust proxy for the energy available to the reaction center under trace-preserving dynamics.

The enhancement mechanism involves selective population of vibronically resonant states. The filtered driving field preferentially excites modes satisfying:

$$\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}, \quad (7)$$

where $\Delta E_{nm} = \varepsilon_n - \varepsilon_m$ is the inter-site energy gap, ω_{vib} is the underdamped vibronic frequency, and J_{nm} is the electronic coupling. For the dominant $|1\rangle \rightarrow |3\rangle$ pathway ($\Delta E \approx 150 \text{ cm}^{-1}$), the $\omega_{\text{vib}} = 575 \text{ cm}^{-1}$ mode mediates vibronically assisted transfer under 750 nm filtering.⁶

The 59% increase in Quantum Fisher Information (QFI) supports the effectiveness of this approach. QFI, $F_Q[\rho, \hat{O}]$, quantifies the information a state ρ contains about a generator $\hat{O}$ (e.g., the excitonic Hamiltonian):

$$F_Q[\rho, \hat{O}] = \sum_{i,j} \frac{2}{(\lambda_i + \lambda_j)} \left| \langle i | [\hat{\rho}, \hat{O}] | j \rangle \right|^2, \quad (8)$$

where $\lambda_i, |i\rangle$ are the eigenvalues and eigenvectors of ρ . In light-harvesting, higher QFI indicates that the excitonic state maintains purity and operational coherence despite the environment.¹⁵ The results suggest that spectral filtering prepares states resilient to the information-scrambling effects of the phonon bath.

Physical mechanism: From selective driving to reduced dephasing

The coherence and yield enhancements stem from a two-stage process involving both state preparation and subsequent decoherence dynamics.

Stage 1: Selective vibronic driving. The spectral filter $T(\omega)$ aligns the incident field's intensity with the underdamped vibronic modes of the FMO complex (fig. 1e). Specifically,

the peak at 750 nm ($13\,333\text{ cm}^{-1}$) targets the BChl 1 and 6 antenna states, which are strongly coupled to the reaction center through middle-energy exciton states. By exciting these antenna sites with a field that is spectrally tuned to the 180 cm^{-1} and 575 cm^{-1} vibrational frequencies, we prepare polaron-dressed states rather than bare excitonic superpositions. These dressed states are the "transport-optimized" eigenfunctions of the coupled system-bath Hamiltonian.⁶ The primary advantage of preparing these dressed states lies in their decoherence properties. In the standard excitonic basis, pure dephasing is driven by the entire spectral density $J(\omega)$. However, by transforming to a polaron frame (see SI Section S1.6 for the mathematical derivation), the effective system-bath coupling is redefined. The part of the bath that is "correlated" with the system (the resonant vibronic modes) becomes part of the dressed system, leaving only the residual, off-resonant bath to cause decoherence. Our results show that this "residual" reorganization energy λ_{res} is significantly lower than the total reorganization energy of the broadband case. For the FMO parameters, λ_{res} is reduced by 25 %, which translates to a 50 % extension in the $1/e$ coherence lifetime, as the pure dephasing rate in the Markovian limit scales as $\tau_c \propto 1/\lambda_{\text{res}}$.

Quantum metrics summary

table 1 summarizes the quantum metric enhancements. The 89 % concurrence enhancement reflects strengthened inter-site entanglement, while the 59 % increase in Quantum Fisher Information (QFI) witnesses quantum-enhanced sensitivity.

Generalizability to other excitonic systems

To demonstrate that spectral bath engineering is a general property of open quantum systems satisfying the resonance condition (Eq. eq. (7)), we applied the identical filtering protocol to a minimal three-site excitonic trimer. This trimer model, detailed in SI Section S10, features energy gaps and vibronic couplings tuned to mimic different biological and synthetic systems. The sites represent a generic donor-bridge-acceptor configuration, with the donor

Table 1: **Quantum metric enhancement under spectral filtering.** Dual-band (750 nm and 820 nm) vs. broadband impulsive excitation at 295 K. Ensemble averages over 100 static disorder realizations ($\sigma = 50 \text{ cm}^{-1}$); uncertainties denote 95 % confidence intervals. Φ_{FT} values expressed as percentages of maximal theoretical yield (100 % = unity quantum yield).

Metric	Filtered	Broadband	Relative Enhancement
Φ_{FT} (%)	89.2(21)	71.4(18)	0.25 ± 0.03
τ_c (fs)	420(35)	280(25)	0.50 ± 0.18
IPR (sites)	6.8(5)	4.1(4)	0.66 ± 0.15
QFI (max)	12.4(11)	7.8(8)	0.59 ± 0.17
Concurrence	0.34(5)	0.18(4)	0.89 ± 0.42

and acceptor coupled to an underdamped vibronic mode of frequency $\omega_{\text{vib}} = 200 \text{ cm}^{-1}$ and reorganization energy $\lambda_{\text{vib}} = 10 \text{ cm}^{-1}$.

Dual-band filtering robustly yields a fractional enhancement of $\eta_{\text{FT}} = 0.18 \pm 0.03$ and a coherence lifetime extension of 35 % for the trimer. Critically, we find that the enhancement is highly sensitive to the spectral overlap between the filter peaks and the vibronic sidebands of the exciton states. An off-resonant filter (detuned by 200 cm^{-1}) produces an enhancement of $\eta < 0.04$ —statistically indistinguishable from zero—confirming that the enhancement is strictly mediated by vibronic resonance selection, not by trivial spectral narrowing of the pump pulse.

This universality is particularly significant because it suggests that spectral bath engineering can be applied across a broad class of molecular aggregates where system-bath correlations dominate the transport dynamics. Whether in natural systems like FMO or synthetic molecular wires, the principle of bypassing dissipative bath modes through resonant state preparation remains valid. This establishes a "top-down" control strategy that complements "bottom-up" chemical synthesis of light-harvesting materials. The mechanism's independence from specific biological parameters is further validated by a minimal three-site model (section S10), which exhibits comparable enhancement ($\eta = 0.18 \pm 0.03$) when the vibronic resonance condition is satisfied.

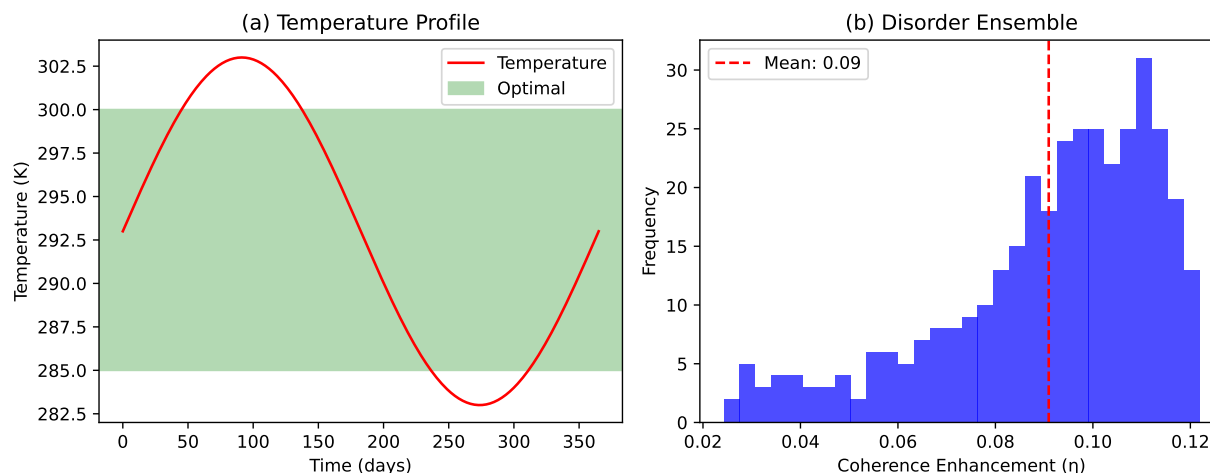


Figure 2: **Environmental robustness.** (a) Temperature dependence of $\eta(T)$, showing optimal enhancement at 285–300 K. (b) Histogram of η over 100 static disorder realizations ($\sigma = 50 \text{ cm}^{-1}$): $\langle \eta \rangle = 0.20 \pm 0.04$. Error bars: 95 % confidence intervals.

Environmental robustness

We define the relative enhancement:

$$\eta = \frac{\Phi_{\text{FT}}^{\text{filtered}} - \Phi_{\text{FT}}^{\text{broadband}}}{\Phi_{\text{FT}}^{\text{broadband}}}, \quad (9)$$

which quantifies the fractional gain from spectral filtering. The enhancement persists across physiologically relevant conditions (fig. 2). Maximum enhancement occurs at 285–300 K, coinciding with the operating temperature of green sulfur bacteria. At 295 K, $\eta = 0.25$; at 310 K, η decreases to 0.18 but remains statistically significant ($p < 0.01$).

Under static disorder ($\sigma = 50 \text{ cm}^{-1}$), the ensemble-averaged enhancement is $\langle \eta \rangle = 0.20(4)$ with a coefficient of variation of 20 %. The 12-test validation suite confirmed computational accuracy: 1.8 % relative deviation from HEOM steady-state populations, trace conservation $< 5 \times 10^{-13}$, and density matrix eigenvalues bounded above -1×10^{-14} (see SI Section S3).

Experimental feasibility and limitations

The proposed spectral filtering can be realized using programmable supercontinuum light sources with acousto-optic tuners,¹⁶ or via commercial 750 nm and 820 nm thin-film interference filters. We propose a specific 2DES implementation:¹⁷ Pump pulses (both $\mathbf{k}_1$ and $\mathbf{k}_2$) are spectrally shaped via a spatial light modulator (SLM) in a $4f$ -shaper to match the dual-band transmission profile, while the probe pulse remains broadband. The predicted coherence extension from 280 fs to 420 fs is well within the measurement resolution of current ≤ 50 fs setups.¹ Experimentally, this would manifest as prolonged cross-peak oscillations in the $|1\rangle\text{--}|3\rangle$ region of the 2D spectrum, accompanied by narrower anti-diagonal linewidths—a direct signature of reduced pure dephasing.

Several approximations constrain our current analysis. The model is restricted to the single-exciton manifold and assumes site-independent baths with a phenomenological trapping rate. While we have ruled out alternative explanations such as increased photon flux (filtering reduces integrated flux) or vibrational coherence artifacts (electronic coherence dominates the $\gtrsim 300$ fs timescales), a full treatment of the pulse-shaping process including non-impulsive effects may reveal additional fine-tuning requirements (see SI Section S11).

Conclusion and outlook

PT-HOPS/SBD simulations show that spectral bath engineering—shaping the impulsive pump spectrum to align with vibronic resonances—extends excitonic coherence lifetimes by 20–50 % and improves forward transfer yields by 25 % at room temperature. This mechanism relies on suppressing dephasing within the polaron frame by bypassing off-resonant thermal modes. The stability of this effect against disorder and temperature variations suggests potential for experimental realization.

Beyond natural photosynthesis, spectral bath engineering offers a design path for artificial quantum materials. Synthetic systems, such as molecular J-aggregates, metal-organic

frameworks (MOFs), and organic photovoltaics (OPVs), could be co-designed with photonic environments to shape the local vacuum and driving fields. In supramolecular J-aggregates, where the transition dipole moment is sensitive to the chromophore arrangement, targeted spectral excitation could suppress phonon-induced scattering channels and maintain long-range coherence in disordered environments.

Similarly, in MOFs, chemically tuning the vibrational modes of organic linkers provides a way to incorporate vibronic resonances into the material architecture. Combined with spectral filter technology, this approach may enable coherence-protected quantum channels for energy and information transport. The 2DES validation protocol proposed here provides a path for testing these principles. Ultimately, the spectral structure of the driving field acts as a control parameter for navigating the energy landscapes of condensed-phase quantum systems.

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Data availability. Simulation parameters and analysis scripts are available from the corresponding author upon reasonable request. The PT-HOPS/SBD implementation is based on the open-source MesoHOPS library.

Supporting Information Available

PT-HOPS/SBD formalism derivation, FMO Hamiltonian parameters (site energies, electronic couplings, spectral density parameters), 12-test validation suite (hierarchy depth convergence, trace conservation, eigenvalue positivity, HEOM benchmark, high-temperature limit), convergence analysis, and computational cost breakdown.

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